Automatic Keyphrase Extraction: A Survey of the State of the Art

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Automatic Keyphrase Extraction (AKE)

"The automatic selection of important and topical phrases from the body of a document." [Turney, 2000]

Example (keyphrases highlighted)

Canadian Ben Johnson left the Olympics today "in a complete state of shock," accused of cheating with drugs in the world's fastest 100-meter dash and stripped of his gold medal. The prize went to American Carl Lewis. Many athletes accepted the accusation that Johnson used a muscle-building but illegal anabolic steroid called stanozolol as confirmation of what they said they know has been going on in track and field. Two tests of Johnson's urine sample proved positive and his denials of drug use were rejected today. "This is a blow for the Olympic Games and the Olympic movement," said International Olympic Committee President Juan Antonio Samaranch.

Applications

Text summarization, text categorization, opinion mining, document indexing

Goal

Survey the state of the art, examine major sources of errors and challenges

Common Evaluation Datasets

Source	Dataset	#docs	#tokens/doc	#keys/doc
Paper abstracts	Inspec (Hulth, 2003)	2,000	<200	10
Scientific papers	SemEval-2010 (Kim et al., 2010b)	284	>5K	15
Technical reports	NZDL (Witten et al., 1999)	1,800	-	-
News articles	DUC-2001 (Wan and Xiao, 2008b)	308	900	8
Webpages	Yih et al. (2006)	828	-	-
Meeting transcripts	ICSI (Liu et al., 2009a)	161	1.6K	4
Emails	Enron corpus (Dredze et al., 2008)	14,659	-	-
Live chats	Library of Congress (Kim and Baldwin, 2012)	15	-	10

Complicating Factors

- Long texts with many candidate keyphrases
- Structural inconsistency (e.g., web pages)
- Topic change in conversational text (e.g., chats)
- Uncorrelated topics in informal text (e.g., emails)

AKE Algorithms: Two steps

Step 1: Selecting candidate keyphrases heuristically

- Remove stop words
- Allow words with certain PoS tags
- Use n-grams from Wikipedia titles
- Use n-grams or noun phrases

Step 2: Finding keyphrases from the candidate set using supervised or unsupervised approaches

Supervised Approaches

- Recast AKE as a binary classification problem (Frank et al., 1999)
- Train a classifier to determine whether a candidate is a keyphrase
- **Weakness**: classifier doesn't compare candidate keyphrases
- **Solution**: ranking (Jiang et al., 2009): train a ranker to rank candidates

Supervised Approaches (Cont')

- **Features**

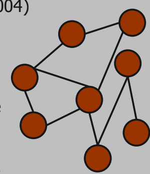
1. Within-Collection: computed based on labeled training documents. Three types:
   - Statistical: KEA's (Witten et al., 1999) feature set - tf*idf, distance, keyphraseness
   - Structural: encode how candidates are located in a document
   - Syntactic: PoS and suffix sequence; ineffective (Yih et al., 2006; Kim and Kan, 2009)
2. External Resource-Based: computed from external knowledge bases (e.g., Wikipedia)
   - Wikipedia-based keyphraseness: does the candidate often appear as a link in Wiki?
   - Query log: is the candidate used as a search query?
   - Coherence: is the candidate semantically related to other candidates?

Unsupervised Approaches

1. Graph-Based Ranking: Example - TextRank (Mihalcea and Tarau, 2004)

- Build a graph from an input document
  - Nodes: candidate keyphrases
  - Edge weights: relatedness between two candidates
- A node's score is defined recursively based on its edges and the scores of the neighboring nodes
- Top-scored nodes are used to form keyphrases

Weakness: The keyphrases may not cover all the document topics



Each node is a word

2. Topic-Based Clustering:

Motivation: The keyphrases should cover all the topics. Three representative systems:

a. KeyCluster (Liu et al.,2009b): Cluster the candidates; one cluster per topic. Select the candidates close to the centroid of each cluster as keyphrases.
Weakness: Each topic (i.e., cluster) is considered equally important

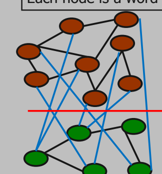
b. Topical PageRank (TPR) (Liu et al., 2010):
- Run TextRank multiple times for a document, once for each of its topics
- Score of a candidate depends on its score for each topic and the topic's importance

c. CommunityCluster (Grineva et al., 2009):
- Like TPR, CommunityCluster gives more weight to more important topics
- Unlike TPR, it extracts all candidate keyphrases from an important topic

3. Simultaneous Learning (Wan et al., 2007):
Perform AKE and text summarization simultaneously

- Build three graphs for the sentences (S) and the words (W)
  - a S-S graph, a W-W graph, and a bipartite S-W graph
  - Edge weight: relatedness between two nodes
- Apply an iterative reinforcement algorithm to rank the nodes

Each node is a word



Each node is a sentence

4. Language Modeling (LMA) (Tomokiyo and Hurst, 2003):

- Score of a candidate = phraseness + informativeness
- Phraseness: how likely a word sequence is a phrase
- Informativeness: how well a word sequence captures the idea of a document
- Compute phraseness and informativeness using language models built on two corpora
  - Foreground corpus includes documents from which keyphrases are to be extracted
  - Background corpus encodes general knowledge about the world (e.g., the Web)

Evaluation

- Exact match computed using precision (P), recall (R), and F-score (F)
- MT-based metrics rewarding partial matches: BLEU, METEOR, NIST, and ROUGE
- Metrics rewarding better ranking of keyphrases: binary preference measure (Bpref), mean reciprocal rank (MRR), and R-precision

The State of the Art

Dataset	Approach and System [Supervised?]	Score		
		P	R	F
Abstracts (*Inspec*)	Topic clustering (Liu et al., 2009b) [No]	35.0	66.0	45.7
Blogs	Topic community detection (Grineva et al., 2009) [No]	35.1	61.5	44.7
News (DUC-2001)	Graph-based ranking for extended neighborhood (Wan and Xiao, 2008b) [No]	28.8	35.4	31.7
Papers (SemEval-2010)	Statistical, semantic, and distributional features (Lopez and Romary, 2010) [Yes]	27.2	27.8	27.5

Error Analysis

Setup

- Four datasets
  - Inspec abstracts, DUC-2001 news articles, scientific papers, and meeting transcripts
- Four systems
  - tf*idf, two unsupervised (Wan and Xiao, 2008b; Liu et al., 2009b), and one supervised (Medelyan et al., 2009)
- Randomly selected 25 documents from each dataset for error analysis

Error Types

1. Overgeneration [Precision error; 28–37%; **Example**: Olympic movement]
   - occurs when a system incorrectly predicts a candidate as a keyphrase because it contains a word that appears frequently in the associated document
2. Infrequency [Recall error; 24–27%; **Example**: 100-meter dash, gold medal]
   - occurs when a system fails to identify a keyphrase owing to its infrequent presence
3. Redundancy [Precision error; 8–12%; **Example**: Olympics and Olympic games]
   - occurs when a system outputs multiple semantically equivalent candidates
4. Evaluation [Recall error; 7-10%; **Example**: Olympics and Olympic games]
   - occurs when scoring programs cannot recognize semantically equivalent predictions

Recommendation: Use background knowledge from external lexical resources
- **Example**: Freebase (40+ million topics, 70+ domains with type definitions

1. Overgeneration
   - No topic named Olympic movement in the Olympic domain
2. Infrequency
   - Exploit relationship between candidates
     - Relate 100-meter dash to Ben Johnson via the Sports domain
     - Relate gold medal" to Ben Johnson via the Olympics domain
3. Redundancy
   - Both Olympics and Olympic games are mapped to Olympic games topic
4. Evaluation
   - Identify semantically equivalent keyphrases during manual labeling
   - Designing scoring programs to identify semantic equivalences

Challenges

Incorporate background knowledge; handle long documents; improve evaluation schemes